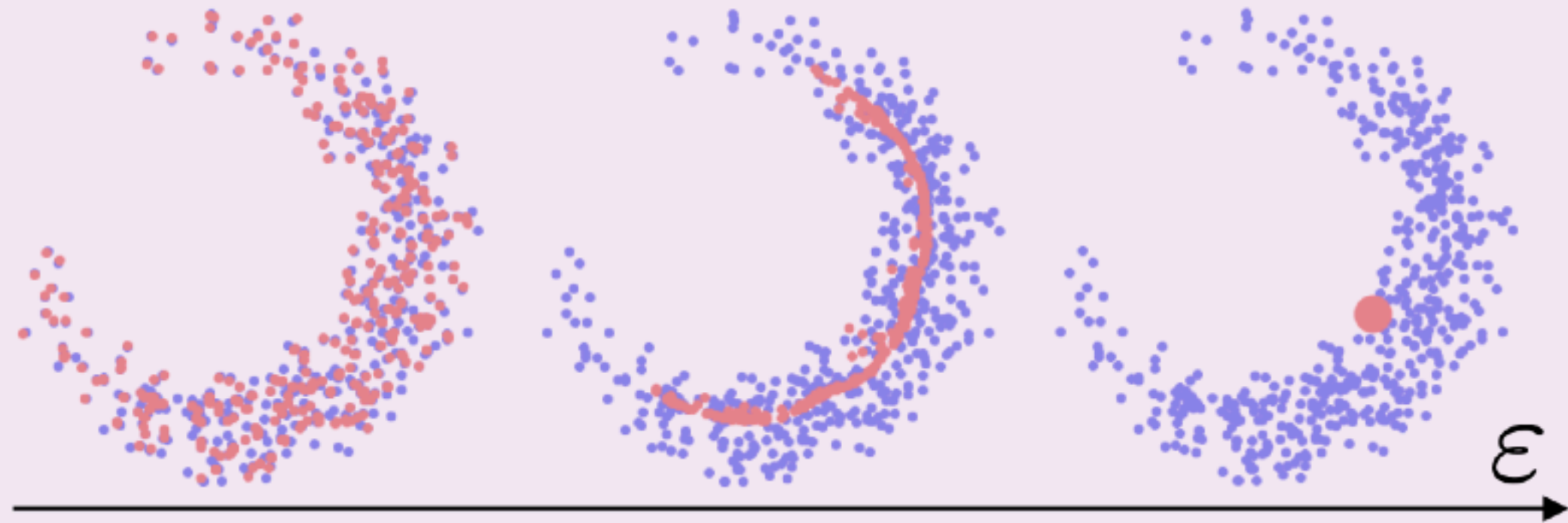


EOT is biased

Problem: $W_\varepsilon(\alpha, \alpha) \neq 0$

$$\min_{\alpha} W_{\varepsilon, p}^p(\alpha, \beta)$$

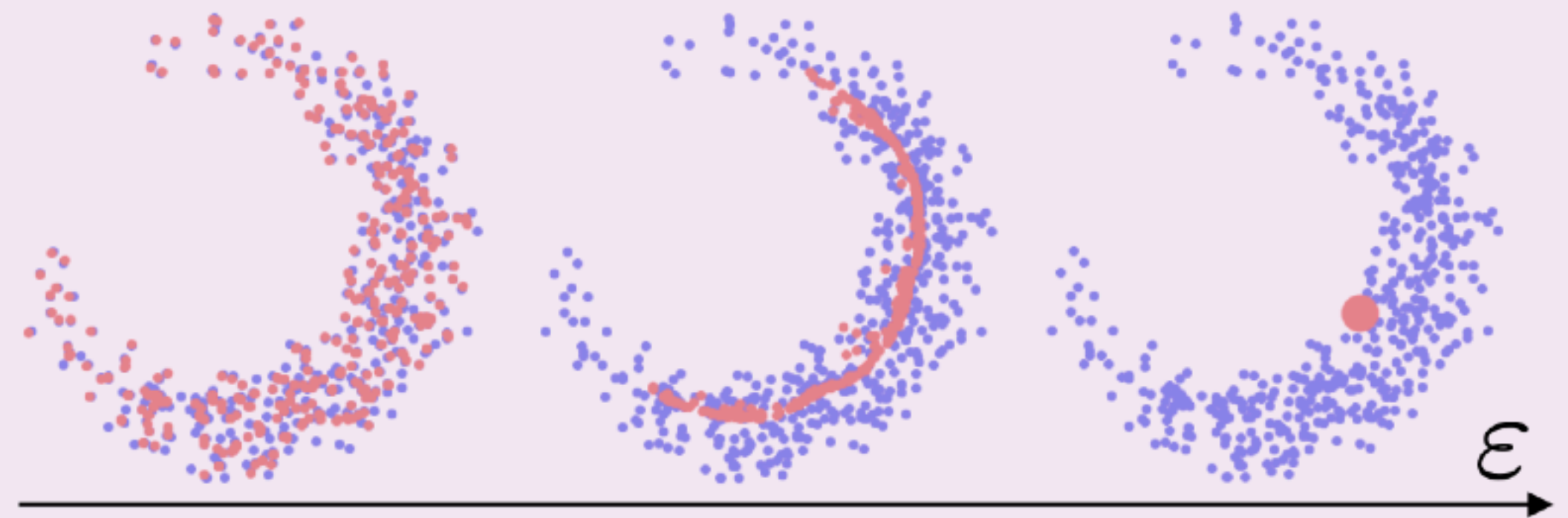


Sinkhorn Divergence Motivation

- Issue with entropic cost: $W_\varepsilon(\alpha, \alpha) > 0$, i.e. a distribution has nonzero distance to itself.
- This means **EOT is biased** as a dissimilarity measure.
- Consequence: can't be used directly as a “distance” in applications.

Problem: $W_\varepsilon(\alpha, \alpha) \neq 0$

$$\min_{\alpha} W_{\varepsilon, p}^p(\alpha, \beta)$$



Sinkhorn Divergence